

Tutorial 7 - SPM (wk12 -done)

Question 1

In the context of version management, identify and explain TWO (2) features of Distributed Version Control and Centralized Version Control Systems in software development.

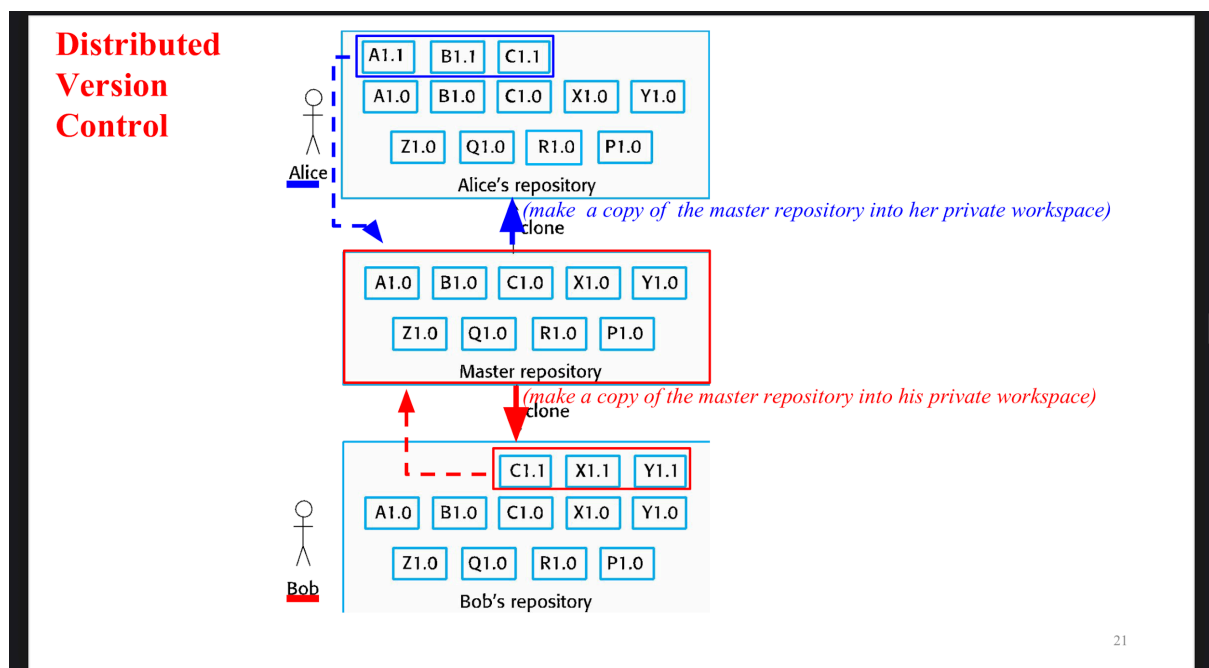
Distributed Version Control

1. Full local clone and offline commits

Each developer clones the entire repository to their machine, can commit locally without network access, and later push to the shared repository.

2. Natural backup and easy integration flows

Because every clone is a complete copy, the project can be restored from developers' machines if the server is corrupted.



Centralized Version Control

1. Single master repository with check-out/check-in workflow

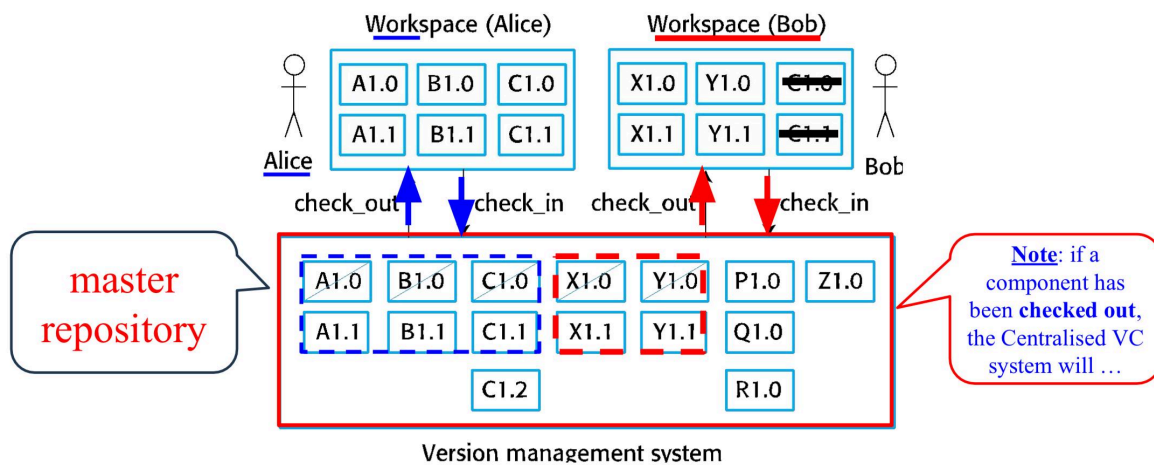
There's one authoritative repository, developers check out files, edit in a private workspace, and check in updates.

2. Always-on network dependency and single point of control

Work typically depends on connectivity to the central server, which simplifies control but creates a single operational bottleneck/failure point compared with DVCS.

Centralized Version Control (An example of a widely used CVC is **Subversion**)

7.2 Version Management



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Question 2

Question 2

System release is a crucial phase in the software development lifecycle where the developed software is made available to users. Describe TWO (2) factors that influence system release planning.

Factors Influencing System Release

7.5 Release Management

Factor	Description
Competition	For mass-market software, when a competing product has introduced new features market share may be lost if a new system release is not provided to your existing customers.
Platform changes	When a new version of the OS platform is released, you may have to create a new release of the software application for the updated OS (E.g: Microsoft upgraded Windows 10 to Windows 11. Your game software also must be updated to make it compatible with Windows 11. Else old game software will not function smoothly)
Technical quality of the system	When there are serious system faults which affect many customers, it may be necessary to issue a fault repair release . Minor system faults may be repaired by issuing patches (usually distributed over the Internet) that can be applied to the current release of the system.
Marketing requirements	The marketing department of an organization have made a commitment for releases to be available at a particular date.

1. Competition

- If another company releases new features, our system also needs a new release.
- Otherwise, we may lose customers because they will move to the

competitor's product.

2. Platform changes

- When a new version of an operating system (like Windows 11) is released, our software must also be updated.
- If we don't update, the old version may not work properly on the new OS.

Question 3

One of the change management activities is to approve the change requests from stakeholders. Discuss ONE(1) factor in change analysis that influences whether the change request should be approved or not.

The benefits of the change

- The consequences of not making the change
- The costs of making the change
- The number of users affected by the change
- The product release cycle